13.6 Errors That Add Up

WORKSHEET

The question

STEP II 1993, Question 15

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/93-s2-q15

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps.

Ask yourself If a number's fractional part U is uniform on $[0,1)$, what is the error when it is truncated? When it is rounded? What are the mean and variance of each?

STAGE 2 The probability of a large error

Ask yourself For LEP, what is $\mathrm{P}(|E| > 15)$ with $E \sim \mathrm{N}(0, 125)$? For VOZ, where is the error almost certainly?

STAGE 3 How many additions?

Ask yourself After n additions, LEP's error has standard deviation $\sqrt{n/12}$. When is $\mathrm{P}(|E| < 10) \geq 0.9$?

Try this next

1. **VOZ corrected.** If VOZ adds $\frac{n}{2}$ to its answer at the end, what is the distribution of its error? Is it now as good as LEP?

2. **Rounding to one decimal place.** If numbers are rounded to one decimal place, with uniformly distributed discarded digits, find the standard deviation of the total error in adding 10 000 numbers.

3. **The Vancouver index.** If the index was recalculated 3000 times a day and each truncation lost on average 0.0005, how much would it lose in a year of 250 trading days?

4. **The exact distribution.** For the sum of two rounding errors, find the exact density (a triangle). For three, sketch it. How quickly does it start to look normal?

Further reading

- **Nicholas Higham, *Accuracy and Stability of Numerical Algorithms*** (SIAM). The authoritative book on rounding errors; the opening chapter is accessible.
- **David Goldberg, "What every computer scientist should know about floating-point arithmetic"**, *ACM Computing Surveys* 23 (1991). A classic, free online.
- **Tim Harford, *How to Make the World Add Up*** (Bridge Street), on systematic errors in

data.

The full guided solution, and what the question leads to at university, is in Section 13.6:
https://beyondstep.keithmcnulty.org/ch13/errors-add-up.html